Developmental Assessmet

* Nomal development.
* Delayed development.
* Red Flags in development. 

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2019 - 2018
"Pediatric Specialist"
" Arab board"

(العوامل بتاعة نمو وتطور Develop)

* factors affecting
① Familial
②- Personal
③- GA pT- Term
④- nutritional status
⑤ Health status
⑥ Environmental

Developmental Assessment

DeF :- is maturation of organs & systems & gaining skills & ability of adaptation & Assuming responsibility

* Development pattern ⇒ is Cephalocaudal.
& Proximo - distal.

Consist of

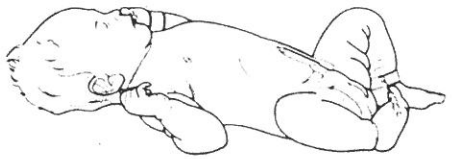
① Gross motor development
② Fine motor development & Vision
③ Social development
④ Hearing, Speech & Language development

Note :- ① Common Age in Exam (4m → 18m)
② 1st is General inspection
- ill - well - oriented or alert - Posture.
- Dysmorphic Features. - Surrounding
③- Don't go systematic.
④- Don't give specific age (Range eg 10-12m)
⑤- Developmental assessment >5y Don't By "school performance"

دعيبة الرحمن الجبلي

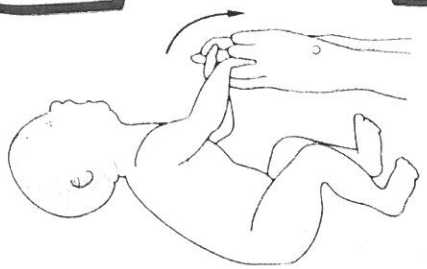
Gross motor development (median ages)

newborn



Limbs flexed, symmetrical postures

newborn

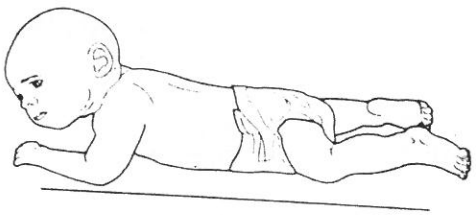


Marked head lag on pulling up

3 months

Head control

6–8 weeks



Raises head to 45°

6 months

1. sits with support ē Round back
2. Rolls Front-to back.
3. Lefts chest up on extented arms.



at 8 months • sits Back (shown)
• Rolls back to Front.
• parachute (7m)

8–9 months



Crawling

10 months



Walks around furniture

12 months



Walking unsteadily, broad gait, hands apart

15 months



Walks alone steadily

Pattern of child development

• 18 months.

- ⊙ Runs well
- ⊙ walks independently.
- ⊙ Creeps down stairs. & walks up stairs (2Feet)

• 2 years.

- ⊙ kicks ball.
- ⊙ Walks down stairs holding rail, both Feet on each step.
- ⊙ Jumps e both Feet From Floor

• 3 years.

- ⊙ walks up stairs 1 Foot per step, down with 2.
- ⊙ pedals tricycle.
- ⊙ walks on tip toes - Stand breiFly on one Foot (2sec)

• 4 years.

- ⊙ walks up & down stairs 1 Foot per step.
- ⊙ Hop on one Foot.

• 5 years.

- ⊙ Runs on toes.
- ⊙ Catches ball.
- ⊙ pedals bicycle.

< 6m of Age (No of head control)

Prone Position

- at birth — Flat
- 6wk — Head & chin
- 3m — Head & chest & flexed elbow
- 5-6m — Head & chest & extented elbow







ventral Suspension

- at birth — C shap
- 6wk — at same level
- 3m — Head above the level of trunk









Vision

- <6m Follow object → 6wks 90°, 3m 180°
- >4y visual acuity Test

- Pubillary react to light at birth & fixation
- Binocular Vision → at 2years
- Full visual acuity → at 4years
- Sequint <3-6 m is normal

④ b

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جميع الحقوق محفوظة
2021

Vision and fine motor (median ages)

6 weeks



- Newborn - follows face in midline.
- Follows moving object or face by turning the head.

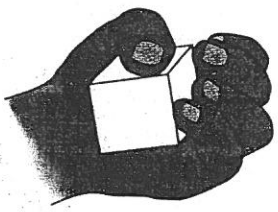
6 wks → Fixes & Follows to 90°.

3–4 months



- Reaches out for toys
- Fix and Follows to 180°

6 months



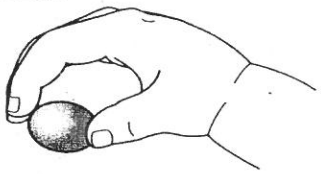
- Palmar grasp
- Transfers objects.
- mouths objects - finger feed

7 months



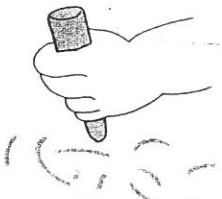
- Transfers toys from one hand to another
- shakes rattle.

10–12 months



- immature pincer grip (9-10m)
- Mature pincer grip (12 m)
- points & index finger.

18 months



- Turns pages of books (18m)
- Makes marks with a crayon
- Hand preference (18-24 m)

14 months–4 years



Tower of three (18 months)

Tower of six (2 years)

Tower of eight or a train with four bricks ($2^1/_2$ years)

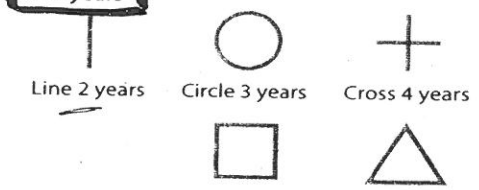


Bridge (from a model) 3 years



Steps (after demonstration) 4 years

2–5 years



Line 2 years

Circle 3 years

Cross 4 years

Square $4^1/_2$ years

Triangle <5 years

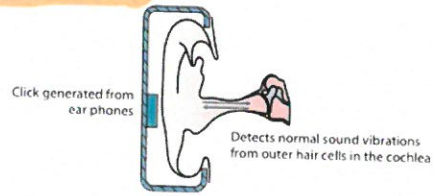
- Ability to draw without seeing how it is done.
- Can copy (draw after seeing it done) 6 months earlier.

Vision and fine motor skills (median ages).

Hearing tests

Hearing screening of newborn infants

a Otoacoustic emission (OEA)

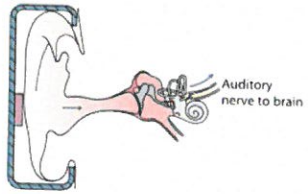


Advantages:
- simple and quick to perform, though is affected by ambient noise

Disadvantages:
- misses auditory neuropathy as function of auditory nerve or brain not tested
- relatively high false-positive rate in first 24 hours after birth as vernix or amniotic fluid are still in ear canal
- not a test of hearing but a test of cochlear function

b Auditory brainstem response (ABR)

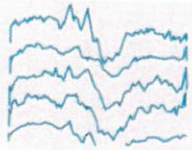




Auditory stimulus via earphones

Signal via ear and auditory nerve to brain

EEG waveforms – computerised analysis determines if normal or abnormal



Advantages:
- screens hearing pathway from ear to brainstem
- low false-positive rate

Disadvantages:
- affected by movement, so infants need to be asleep or very quiet, so time consuming
- complex computerised equipment, but is mobile
- requires electrodes applied to infant's head, which parents may dislike

Figure 3.9 Universal neonatal hearing screening is usually performed using (a) otoacoustic emission testing or (b) auditory brainstem response audiometry.

Hearing distraction Test

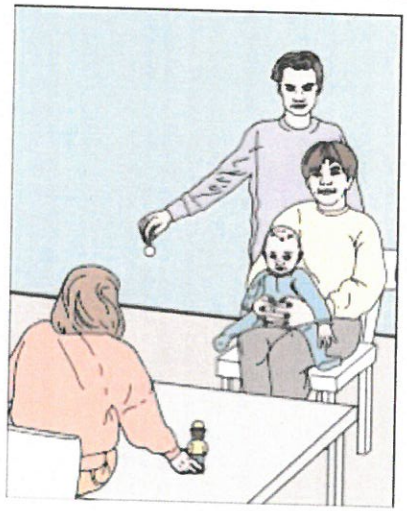
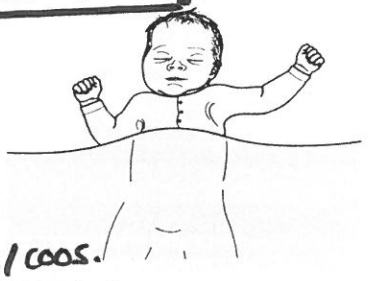


Figure 3.10 Distraction hearing test. The test is hard to perform reliably as babies with hearing difficulties learn to compensate by using shadows, smells and guesswork to locate the presenter. The test must be done by well-trained professionals

Hearing, speech and language (median ages)

NEWBORN – 6 wks



- cries / coos.
- Startles to loud noises

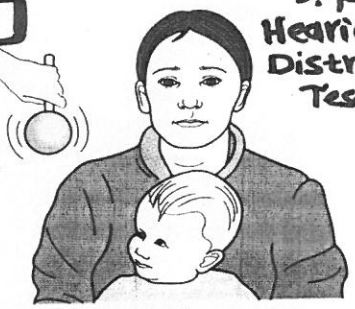
3–4 MONTHS



- Turn to sound.
- Vocalises alone or when spoken to, coos and laughs

7 MONTHS

حكم جدا
Hearing Distraction Test



- Turns to soft sounds out of sight

7–10 MONTHS



- At 7 months, sounds used indiscriminately.
- At 10 months, sounds used discriminately to parents

12 MONTHS



- Two to three words other than 'dada' or 'mama'
- knows name.

18 MONTHS



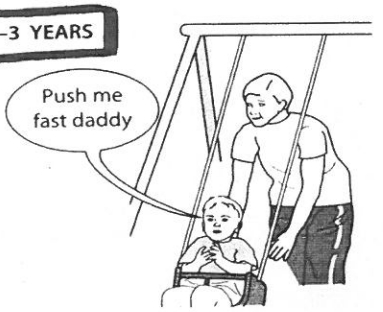
- 6–10 words. Shows two parts of the body

20–24 MONTHS



- Uses two or more words to make simple phrases

$2\frac{1}{2}$–3 YEARS



- Talks constantly in 3–4 word sentences
- knows sex.
- Recognizes colours. [3yr]
- give 1st & last name [3yr]

4 years

- Counts to 10 or more

5 years

- asks "How" & "when" questions

Pattern of child development

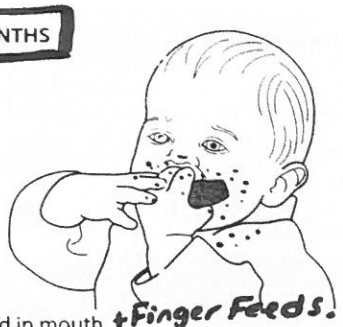
Social, emotional and behavioural development (median ages)

6 WEEKS → 3m



a
- Smiles responsively (6wks) – Recognize mother face
- Laughs. (3m)

6–8 MONTHS



- Puts food in mouth + Finger Feeds.
- Stranger anxiety (8m)

10–12 MONTHS



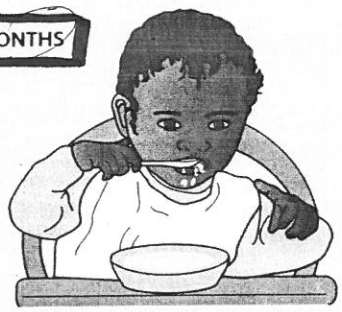
c
- Waves bye-bye, plays peek-a-boo

12 MONTHS



d
- Drinks from a cup with two hands
- Finds hidden object.

18 MONTHS



e
- Holds spoon and gets food safely to mouth
- Symbolic plays alone.

→24 MONTHS



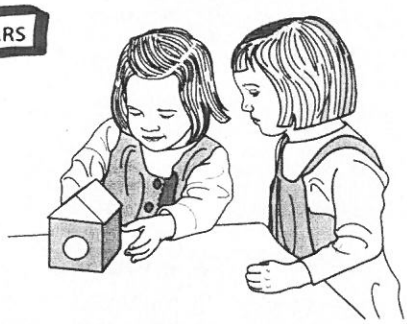
f
- Feeds with Fork.
- Symbolic play

2 YEARS



g
- Dry by day.
- Pulls off some clothing

3 YEARS



h
- Parallel play. Interactive play evolving. Takes turn
- washes hands & brushes teeth.

4 years
- Able to undress
- points to color on request [Red - Blue - green]

5years
- Uses knife & Fork.

Causes of developmental Delay.

Developmental Quotient (DQ) = $\frac{\text{Developmental Age}}{\text{Chronological Age}} \times 100$

- ≥ 85 Normal
- 71-84 mild to mod delay
- ≤ 70% sever Delay

Global developmental Delay (most common)	Delayed speech / language	Delayed motor development
1. HIE.	Central :-	.Central :-
2. Cerebral malFormation	1. Autism	1. cerebral palsy.
3. Chromosomal disorder	2. Hearing loss.	2. Hemiplagia.
4. TORCH infections.	3. Isolated speech Delay.	3. Visual impiarment
5. metabolic disease (IEM) NF-TS	Enviromental :-	. Systemic disorders
6. Toxin exposure.	1. malnutrition	1. Hypo T_3T_4
7. Hypo T_3T_4 - PKU.	2. lack of practice	. Enviromental :-
8. Hypoglycemia (sever & Recurrent)		1. lack of practile
		2. malnutrition

Mental Retardation

incomplete or insufficient mental development of mental capacity (WHO)

$IQ = \frac{\text{mental Age}}{\text{chronological Age}} \times 100$.

[Average IQ 90-110]

- IQ < 70 ⇒ MR
  - 50-70 ⇒ mild MR
  - 35-50 ⇒ mod MR
  - 20-35 ⇒ sever MR
  - <20 ⇒ profound MR.

- Causes of MR :-
  1. Prenatal → 1. Chromosomal Trisomy
     - 2. TORCH.
     - 3. IEM PKU.
     - 4. Drugs - irridiation
  2. perinatal → 1. prematurity
     - 2. HIE.
  3. postnatal → 1. meningitis - Encephl
     - 2. ↓ T_3T_4 - glucodemia
     - 3. kirnicturus.
  4. Idiopathic.

(8)

"Causes of Hearing loss"

Conductive	sensorineural
[1] Otitis media & effusion (glue ear)	[1] Genetic.
[2] Cleft palate.	[2] TORCH.
[3] Down Syndrome.	[3] prematurity.
[4] Pierre Robin Syndrome.	[4]. HIE.
[5] Mid-Facial Hypoplasia	[5]. Kirnicterus.
[6] wax (only rarely cause hearing loss	[6]. meningitis-Encephalitis
↓	[7]. Drugs (Gentamycin)
	[8]. NDD. Head injury.
	↓
Rx Conservative, amplification or surgery	Rx Amplification or cochlear implant

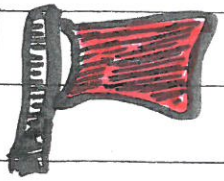
Ix 1. visual assessment
2. Hearing Test
3. • CT & MRI Brain
• TFT
• chromosomal study

Clinic

Rx Multidisciplinary Team
1. speech therapist
2. Hearing ENT
3. visual management
4. Rehabilitation
5. physiotherapy

***

Illustrated Textbook of paediatrics



Red Flags in development

Sign MCQ

* maternal concern. * Regression of previously acquired milestones.

① Red Flags at (2) months of age :-

1. Does not respond to loud sounds.
2. " " Smile.
3. NO visual Fixation
4. lack of sustained visual interst.

② Red Flags at (6) months of age.

1. No head control. "Floppy"
2. Does not respond to sounds around them.
3. D" " make vowel sound "Bubbling" (ah-eh-oh)

③ Red Flags at (9) months of age :-

1. Does not sit ē support.
2. inability to localize the sound.
3. persistant premitive Reflexes (77)

④ Red Flags at (12) months of age.

1. Can't stand ē support. - persistant mouthing & rolling } 12m
2. Does not say single word like (Baba, mama)
3. Hand dominance (<18m) (Handness <6.12m)

⑤ Red Flags at (18) months of age :-

1. Can't walk.
2. loses skills they once had.
3. Does not know what Familiar things are for.

⑥ Red Flags at (24) months of age

1. Does not use two word phrases.
2. Does " wake steadily

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Test Yourself







د.عبدالرحمن الجعكي

Developmental Age ???









Fig. 5.44 6 months Immature pal mar grasp of cube

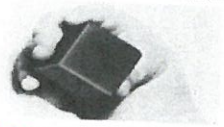
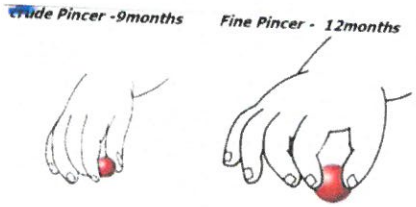


Fig. 5.45 8 months Grasp, intermediate stage



Fig. 5.46 1 year Mature grasp of cube





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